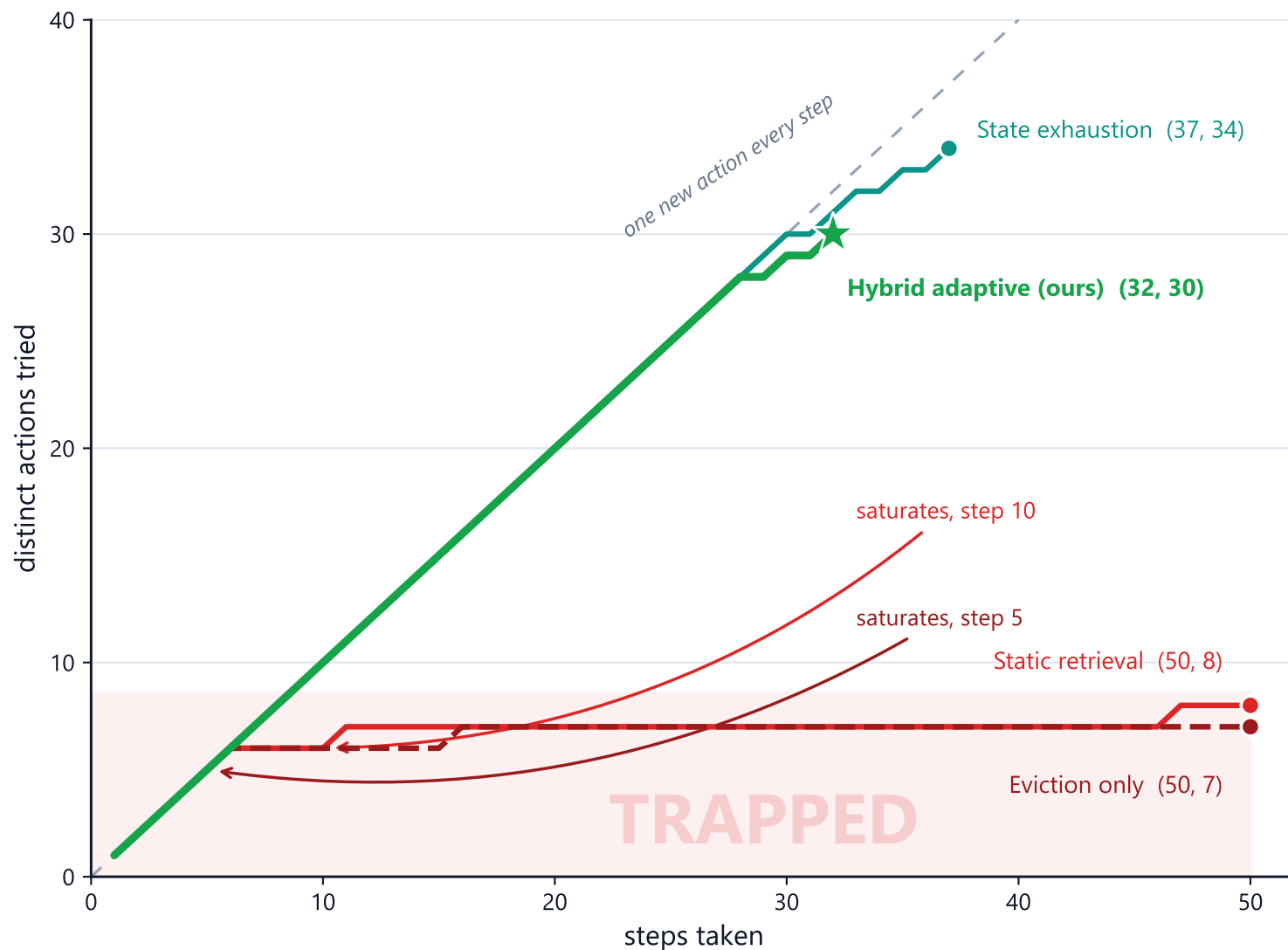


The retrieval trap is a saturation phenomenon

ALFWorld, out-of-distribution layout. Distinct actions tried vs. steps taken.



One seed per variant, 50-step budget, identical model, tools and base prompt. Curves are the raw action logs.

Same start, different fate

All four agents open with nearly the same six actions. The prior is good advice until it is not.

Eviction alone is worse

Dropping a failed prior leaves nothing behind: it saturates at step 5, sooner than the baseline it fixes.

Replace, do not just remove

Eviction plus a state-exhaustion fallback keeps searching: 30 distinct actions in 32 steps, task solved.